MADHU KRUPESH

Chicago, IL | 773-869-4782 | madhukrupesh@gmail.com | www.linkedin.com/in/madhukrupesh

EDUCATION

Illinois Institute of Technology, Chicago, IL — JAN 2024 - EXPECTED DEC 2025

Master of Science, VLSI, and Microelectronics, Computer Hardware Engineering

Alva's Institute of Engineering and Technology, Mangalore, KA, IN — AUG 2019 - JUL 2023

Bachelor of Engineering, Electronics and Communication Engineering

SKILLS

- **EDA Tools**: Cadence Virtuoso | Synopsys Design Compiler | Xilinx Vivado | Intel Quartus Prime | ModelSim | SimVision
- **Simulation & Analysis**: HSPICE | PSPICE | SpectreRF | MATLAB | Synopsys Primetime | OrCAD | TCAD | KiCAD
- **VLSI Design**: RTL-to-GDSII | STA | CDC | Low-Power Design | SoC Verification | FPGA/ASIC Design | Analog/RF IC Design
- **Programming & Scripting**: Verilog | System Verilog | UVM | C/C++ | Python | Embedded C | TCL | Perl | Java | VHDL | Shell Scripting
- **Emerging Technologies**: RISC-V SoC | Hardware Security | DSP/AI Accelerators | High-Level Synthesis | Machine Learning (PyTorch) |
- **Protocols**: AMBA (AXI, AHB, APB) | SPI | I2C | UART | PCIe | Ethernet

WORK EXPERIENCE

ENGINEER ASSISTANT

Sigenics, Inc. Chicago, IL, USA — MAR 2025 - PRESENT

- Directed analog and mixed-signal IC projects, migrating schematics in Cadence Virtuoso, validating transistor-level circuits, and aligning layouts with silicon process design kits for fabrication readiness.
- Streamlined verification tasks by coordinating testbench creation, regression tracking, and debug workflows, accelerating tape-out schedules and reducing runtime for custom integrated circuits.

DESIGN VERIFICATION ENGINEER

Proxelera Pvt. Ltd., Mysore, KA, IN — FEB 2023 – JAN 2024

- Assessed 30+ SoC and IP-level designs using SystemVerilog, Verilog, and UVM, closing functional coverage across AMBA, I2C, SPI, and UART protocols to strengthen design reliability.
- Constructed an out-of-order CPU simulator in C++ and delivered an AHB-to-APB bridge, ensuring functional correctness through exhaustive regression, improving throughput and scalability by 10%.

ENGINEER INTERN

Q-Spiders, Bangalore, KA, IN — APR 2022 – JAN 2023

- Enhanced 10+ testbench environments in SystemVerilog, C++, and Python with UVM frameworks (ModelSim, Synopsys VCS), reducing simulation overhead by 20% and enabling efficient SoC bring-up.
- Delivered multi-language verification projects (C/C++, Python, Verilog/SystemVerilog), refining debug automation and advancing hardware-software co-design, FPGA prototyping, and mixed-signal workflows with Cadence and Xilinx toolchains.

RESEARCH INTERN - VLSI & HARDWARE DESIGN

Envision Lab, AIET, Mangalore, KA, IN — AUG 2021 – MAR 2022

- Produced 15+ PCB prototypes in KiCad with structured validation, cutting hardware integration issues by 25% and prototype errors by 30%, expediting fabrication cycles and project milestones.
- Validated 5+ digital subsystems in SystemVerilog, deepening ASIC design expertise and improving verification accuracy by 20% through iterative simulation and debug analysis.

PROJECTS

Energy-Efficient VLSI Design and Optimization Framework

- Formulated RTL, gate-level, and transistor-level modules (Shift-Add Multiplier, VLIW pipeline, cache systems) with operand isolation, clock gating, and memory partitioning, achieving ~35% lower power and 20% faster timing in 22nm HSPICE.

Low-Power FinFET Logic and SRAM Design for Advanced VLSI Systems

- Implemented Dual-Vt domino logic and 6T/8T SRAM cells in FinFET, reducing leakage by ~35% and delay by ~20%. Verified resilience via waveform simulations, optimizing power–performance balance for advanced VLSI systems.

High-Performance FPGA Accelerator (VGG11)

- Crafted a systolic-array CNN accelerator with advanced pipelining and tiling, delivering 3× inference speedup and 35% energy efficiency. Confirmed scalability through FPGA prototyping for reliable deployment in AI-driven SoCs.

RTL-to-GDSII CPU Design Flow

- Designed a 32-bit pipelined CPU (ALU, datapath, memory, control) using Cadence and Synopsys EDA, achieving 18% delay, 12% area, and 15% power reduction for tape-out-ready silicon.

RISC-V Hardware-Software Co-Design with AI Acceleration

- Prototyped a RISC-V CPU with integrated AI accelerators, applying hardware-aware scheduling and software-driven tuning to achieve 2.5× CNN inference speedup and 25% lower power across scalable workloads.

AHB-to-APB Bridge Implementation

- Realized an AMBA protocol bridge in Verilog with Synopsys and Cadence flows, boosting data transfer by 10% and SoC scalability by 10%. Verified correctness through simulation and synthesis for robust VLSI deployment.

HONORS AND ACHIEVEMENTS

- Selected as Student Researcher in MEMS Lab, advancing sensor precision by 15% and cutting simulation runtime by 20% using CoventorWare modeling for microsystems.
- Led a 5-member robotics team to Top 10 among 1,200+ entries in IIT Bombay's All-India Robotics Competition (e-Yantra), delivering automation that increased efficiency by 30%.
- Earned IPC Certified Specialist credential in electronics assembly, PCB design standards, and quality assurance, reinforcing manufacturability and reliability across advanced semiconductor design flows.